In pursuit of the 'Unbreakable' Athlete: what is the role of moderating factors and circular causation?

Tim J Gabbett,[1,2] Rasmus Oestergaard Nielsen,[3] Michael Lejbach Bertelsen,[3] Natália Franco Netto Bittencourt,[4,5] Sérgio T Fonseca,[6] Shane Malone,[7,8,9] Merete Møller,[10] Eric Oetter,[11] Evert Verhagen,[12,13] Johann Windt[14,15,16]

The goal for sports medicine practitioners is to develop robust athletes, capable of withstanding high training and competition loads. For sports medicine professionals, understanding the workload–capacity relationship is central to achieving this goal. This editorial discusses how two different methodological frameworks—(1) moderation and (2) circular causation—align to develop physical capacity and injury resilience in athletes.

'THE STRAW THAT BROKE THE CAMEL'S BACK'

The Arab proverb '*the straw that broke the camel's back*' refers to a camel carrying a haystack that was so heavy a single piece of additional straw broke its back. In a sport setting, the '*camel*' is the athlete, the '*load of hay*' represents the maximal workload the athlete can tolerate safely (load capacity), and the '*additional straw*' represents overload resulting in injury (capacity exceeded). The inherent biological qualities of the camel (eg, age, strength and so on) determine its cumulative *straw*-carrying capacity.

Basketball superstar Kobe Bryant had strong training ethic, well-developed physical qualities and was largely injury-free in the early stages of his career. He was a strong 'camel' accustomed to carrying large loads with ease. We speculate that his athletic pursuits in childhood and adolescence contributed to his high load capacity in adulthood. Indeed, weight-bearing physical activity during childhood and early puberty has a positive, and possibly enduring effect on bone strength.[1] Furthermore, after a certain age, the eccentric heart hypertrophic adaptation (thus maximal stroke volume) seems to be fixed, thus becoming a potential limiter in peak aerobic performance.[2] These findings suggest that high volumes of moderate-intensity activity as an adolescent may facilitate greater adaptations as an adult. Despite Bryant's physical qualities, he had an injury-plagued end to his career; he averaged 34% fewer games in his final five seasons than his first fifteen. Is it possible that his strong training ethic and well-developed physical qualities protected him from injury during his career, but age—a moderating factor—made him more vulnerable to injury later in his career?

Age,[3] strength[4] and biomechanical factors[5] moderate the workload–injury relationship. In addition, emotional and lifestyle stressors influence training adaptation.[6] Given multiple factors interact in determining physical capacity, answering the question 'how much training load is too much among athletes with different characteristics?' it is necessary to assess the relevant musculoskeletal system (MSK) variables that determine load tolerance. To increase capability, we must improve the MSK capacity to respond to mechanical energy (load) applied to the kinetic chain.[7] For example, if two basketball players jump the same amount, the player with restricted ankle dorsiflexion range of motion may benefit from specific treatment and exercises to regain ankle range of motion and thus return patellar tendon load back to normal. However, many questions remain unanswered in relation to load capacity: How is a large capacity best developed and maintained? How large should the load capacity be? Can load capacity ever be so large that the athlete becomes unbreakable?

WHAT COMES FIRST—THE ROBUST ATHLETE OR THE HIGH TRAINING LOAD?

Athletes with well-developed physical capabilities (eg, strength, aerobic fitness, speed, repeated-sprint ability) are more robust against injury at higher training loads.[4] The association between physical capability and injury is moderated through increases in structure-specific load capacity.[8] However, when inspecting this relationship between training load, physical capabilities and injury risk, the '*Chicken or Egg*' question arises. To develop and maintain robust physical capabilities, athletes must complete high training loads. However, tolerating high training load and improving physical capabilities require structure-specific load capacities that depend on already having robust physical qualities. In this sense, the workload–capacity relationship forms a chronological 'Chicken and Egg' question best solved through appropriate load management. Appropriately progressed training load improves capacity, which in turn improves the athlete's ability to tolerate further training load.

TWO ANALOGIES TO ALIGN THE METHODOLOGICAL FRAMEWORKS OF MODERATION AND CIRCULAR CAUSATION

The 'Chicken or Egg' analogy is a case of *circular causation*; training load develops physical capacity, but physical capacity is also required to tolerate load. On the other hand, the 'Straw and Camel' analogy refers to *moderators* of the workload–capacity relationship. These analogies carry important implications for both researchers and sports medicine practitioners.

[1]Gabbett Performance Solutions, Brisbane, Queensland, Australia
[2]University of Southern Queensland, Institute for Resilient Regions, Ipswich, Queensland, Australia
[3]Section for Sports Science, Department of Public Health, Aarhus University, Aarhus, Denmark
[4]Physical Therapy Department, Minas Tênis Clube, Belo Horizonte, Brazil
[5]Department of Physical Education, Uni-BH University, Belo Horizonte, Brazil
[6]Graduate Program in Rehabilitation Science, Physical Therapy Department, Universidade Federal de Minas Gerais, Belo Horizonte, Brazil
[7]Research Institute for Sport and Exercise Sciences, Liverpool John Moores University, Liverpool, UK
[8]Gaelic Sport Research Centre, Department of Science, Institute of Technology, Dublin, Ireland
[9]High Performance Unit, Irish Rugby Football Union, Dublin, Ireland
[10]Department of Sports Science and Clinical Biomechanics, University of Southern Denmark, Odense, Denmark
[11]Memphis Grizzlies, Memphis, Tennessee, USA
[12]Department of Public and Occupational Health, Amsterdam Movement Science, Amsterdam UMC, Vrije Universiteit Amsterdam, Amsterdam, The Netherlands
[13]UCT/MRC Research Unit for Exercise Science and Sports Medicine, Department of Human Biology, Faculty of Health Sciences, University of Cape Town, Cape Town, South Africa
[14]Experimental Medicine Program, University of British Columbia, Vancouver, British Columbia, Canada
[15]Sports Medicine, United States Olympic Committee, Colorado Springs, Colorado, USA
[16]United States Coalition for the Prevention of Illness and Injury in Sport, Colorado Springs, Colorado, USA

Correspondence to Professor Tim J Gabbett, Gabbett Performance Solutions, Brisbane, QLD 4011, Australia; tim@gabbettperformance.com.au

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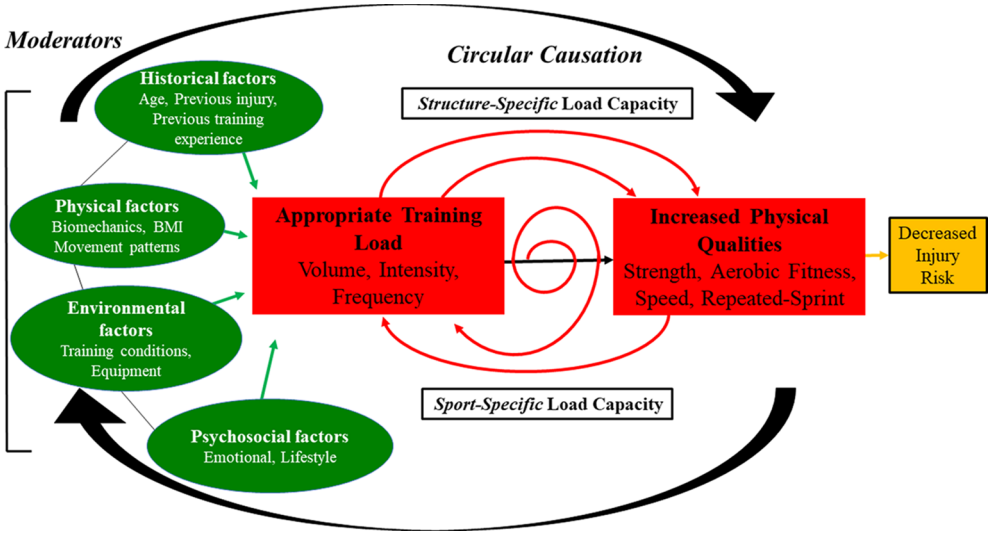

Figure 1 Alignment of two different sports injury frameworks. The *green boxes and arrows* demonstrate how the workload–capacity relationship is *moderated* by historical, physical, psychosocial and environmental factors. The *red boxes and circles* provide an example of *circular causation*. Developing physical qualities requires high training loads, but tolerating high training loads requires well-developed physical qualities. BMI, body mass index.

If the research objective is to identify acceptable training load, researchers need to include total training load and changes in training load (eg, acute:chronic workload ratio) as primary exposures in analyses.[8] Given that injury is multifactorial with numerous moderators of the workload–injury relationship,[9] understanding how these moderators and their relative effects change over time may inform the management strategies employed by sports medicine teams. Considering circular causality, current and past factors influencing load tolerance (eg, previous injury, age, training history, strength, aerobic fitness, sleep, biomechanical and psychological factors) is necessary to determine how to safely increase capacity in athletes (figure 1).

BETTER APPLIED PRACTICE

The 'Chicken or Egg' and 'Straw and Camel' challenges can be solved. Large (perhaps simulated) longitudinal data sets will be required in order to accurately inform the interventions prescribed to a physically fit, but ageing athlete (eg, Kobe Bryant), or one requiring a specific MSK intervention (eg, to increase ankle dorsiflexion range). Computational modelling (where data are generated from simulated athletes)[10] and non-linear time-series analysis (which relies on high-resolution longitudinal data)[9] may help answer these questions. However, any statistical approach must also consider the 'real world' demands of sport; practitioners must collect, interpret and act on data under great time pressure.

To truly understand the workload–capacity relationship requires collaboration among all professionals involved in the training process; strength coaches will 'train', physical therapists will 'treat' and sport scientists will 'test'. Sports biostatisticians and epidemiologists can also play an important role in explaining the different moderators and (circular) causal relationships of training, injury and performance. Ultimately, developing synergistic pathways among all these stakeholders will result in the best applied model for reducing injury risk in pursuit of the unbreakable athlete.

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